Question

N_2O emits a popping sound when it reacts with CS_2. To investigate the products of this reaction, an interest group conducted the following experiment:

A dry and clean test tube was weighed, and its weight was 31.33 g. It was then filled with N_2O under normal pressure. A few drops of CS_2 were added. After the N_2O had reacted almost completely, the generated gas was passed sequentially through a. saturated Na_2CO_3 solution, concentrated sulfuric acid; b. I_2O_5 solid; c. saturated Na_2CO_3 solution, concentrated sulfuric acid. Finally, the volume of the collected gas was similar to the volume of the initial N_2O gas in the test tube. It is known that the mass increase in a. is approximately 4 times the mass decrease in b. After the reaction is complete, the unreacted CS_2 in the test tube is removed and weighed, and the weight is 31.70 g.

It is known that there are 4 kinds of molecules in the products. According to the above information, write the equation for the reaction, paying attention to the actual existing forms of each substance's molecules. Regarding the product of the coefficients in the equation, the correct option is:

A. All other options are incorrect

B. 294912

C. 9

D. 8

E. 521420800

F. 11855241216

G. 300

H. 16384

Answer

Correct Answer: B

Detailed Explanation

First, based on the reactants and experimental phenomena, speculate on the possible products:

The reaction between N_2O and CS_2 is a typical redox reaction. The nitrogen element in N_2O has oxidizing properties, while the sulfur element in CS_2 has reducing properties, possibly producing N_2 and elemental sulfur. The generated gas is passed into a. saturated Na_2CO_3 and concentrated sulfuric acid solution, indicating the generation of acidic gas, possibly SO_2. During the reaction, SO_2 is completely absorbed, and the weight increase corresponds to the mass of SO_2 in the gas: $m(SO_2) = \Delta m(a)$.

CHECKPOINT 1 PTS

The weight increase in a. Na_2CO_3 indicates the absorption of SO_2 in the products

CHECKPOINT 1 PTS

The weight increase in a. corresponds to the mass of SO_2 in the gas

Passed into b. I_2O_5 solid, causing its mass to decrease, indicating that the generated gas also contains reducing gas, and oxygen atoms are transferred to the reducing gas during the reaction with I_2O_5. Considering that SO_2 has been removed by Na_2CO_3, the reducing gas may be CO, and the reaction equation is:

$$I_2O_5 + 5CO \rightarrow I_2 + 5CO_2$$

The weight loss is the decrease of oxygen atoms, and the relationship between the mass of CO in the gas and the weight loss is:

$$m(CO) = 28.01 \times \frac{\Delta m(b)}{16.00}$$

CHECKPOINT 1 PTS

The weight loss in b. indicates that I_2O_5 reacts with CO in the product gas

CHECKPOINT 1 PTS

The weight loss in b. can be used to calculate the mass of CO in the gas

The gas continues to be passed into c. Na_2CO_3 and concentrated sulfuric acid solution, which should remove the gas generated in b., namely CO_2.

CHECKPOINT 1 PTS

Passing into c. can remove the CO_2 generated in b.

The final unreacted gas is similar in volume to N_2O, indicating that the gas product contains inert gas, corresponding to N_2. The volume before and after is the same, indicating that the nitrogen atoms in N_2O are completely converted to N_2.

CHECKPOINT 1 PTS

The gas volume is the same before and after the reaction, indicating that all N_2O is converted to N_2

The mass of the test tube increases after the reaction, indicating that elemental sulfur is formed. The question indicates that there are four types of molecules in the product, which is just right.

CHECKPOINT 1 PTS

The test tube gains weight after the reaction, indicating the formation of elemental sulfur

The unbalanced reaction equation can be written first:

$$xN_2O + zCS_2 \rightarrow xN_2 + ySO_2 + zCO + nS_8$$

The weight increase in a. is approximately 4 times the weight loss in b.: $m(CO) = 28.01 \times \frac{\Delta m(b)}{16.00} = 28.01 \times \frac{\Delta m(a)}{4 \times 16.00} = 0.4377\Delta m(a) = 0.4377m(SO_2)$

$$n(CO) = m(CO)/M(CO) = 0.4377m(SO_2)/M(CO) = 0.4377n(SO_2) \times M(SO_2)/M(CO) = 1.001 \times n(SO_2)$$

That is, the number of moles of SO_2 and CO in the product are the same, $y = z$.

CHECKPOINT 1 PTS

According to the weight relationship between a. and b., it is concluded that the number of moles of SO_2 and CO are the same

In fact, the equation can be solved based on the conservation of the number of atoms:

$$x = 2y + z$$

$$y = z$$

$$2z = y + 8n$$

Reduced to the simplest form, the equation is:

$$24\ N_2O + 8CS_2 \rightarrow 24\ N_2 + 8SO_2 + 8CO + S_8$$

CHECKPOINT 1 PTS

The equation solved according to the conservation of the number of atoms is $24\ N_2O + 8CS_2 \rightarrow 24\ N_2 + 8SO_2 + 8CO + S_8$

The product of the coefficients is 294912, so option B is correct.